

AI Finance Agent Team

Retrieval-Augmented Multi-Agent Decision Support for Market Analysis

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Abstract

AI Finance Agent Team is a multi-agent financial intelligence platform that synthesizes real-time market data, global news context, and analyst sentiment into executive-level analytical reports. The system addresses the persistent gap between natural-language user intent and evidence-backed financial reasoning by implementing a retrieval-first, deterministic multi-agent workflow. Built on Groq and SambaNova LPUs for near-zero-latency inference, the platform delivers structured, source-grounded analysis suitable for analyst-in-the-loop portfolio decision support.

1. Problem Statement

In practical portfolio management, users can access abundant information but lack a reliable mechanism to convert unstructured narrative events into defensible investment decisions. Conventional quantitative models require structured numerical features and therefore underperform for open-ended text queries, whereas generic conversational systems often produce responses without verifiable market grounding.

This creates a persistent gap between natural-language user intent and evidence-backed financial reasoning. The proposed system addresses this gap by implementing a **retrieval-first, multi-agent workflow** that (a) interprets user prompts, (b) acquires live macro and market signals, (c) consolidates evidence into structured context, and (d) generates concise analytical outputs within interactive latency constraints.

2. Data Sources

The platform combines live market telemetry and web intelligence instead of relying on a static local corpus:

Yahoo Finance (yfinance)	Live and historical price series, analyst signals, company fundamentals, recent company-level news
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DuckDuckGo Search API	Broader geopolitical and macroeconomic context relevant to asset movement interpretation
Prompt-Derived Signals	Extracted symbols, entities, and intent tokens from user queries for targeted retrieval and comparison

3. System Architecture

The architecture follows a **Three-Track Pipeline** designed for deterministic, hallucination-resistant financial analysis:

Track 1 — COLLECT (Parallel Ingest)

Parallel retrieval via DuckDuckGo API for unfiltered global news and Yahoo Finance for deep market metrics including price, P/E ratio, sector classification, and earnings data. Both channels execute concurrently to minimize end-to-end latency.

Track 2 — PROCESS (Normalization)

Ticker and entity extraction from natural language using regex and fuzzy alias mapping. Relative date expressions (e.g., 'yesterday', 'Q1 earners') are resolved into absolute timestamps to filter search results and prevent temporal hallucination.

Track 3 — SYNTHESIZE (Multi-LPU Reasoning)

Retrieved evidence is aggregated into a compact, deterministic context window and passed to Groq (Llama-3.3-70B) or SambaNova (Llama-3.1-405B) for structured markdown report generation. Automatic 429 failover ensures 100% uptime between providers.

4. PEAS Framework

Component	Description
Performance (P)	Factual consistency, source coverage, coherence of synthesis, and end-to-end latency. Operational objective is dependable analytical output under interactive response-time constraints.
Environment (E)	Dynamic, stochastic, and partially observable. Market states and news narratives evolve continuously; prompt scope may shift abruptly between company-specific and macro-level perspectives.
Actuators (A)	Multi-agent orchestration combining web retrieval, finance tools, and LLM synthesis; interactive dashboard modules for query execution, comparison, and diagnostics; structured answer generation for portfolio-oriented decision support.
Sensors (S)	Natural-language query stream from end users; ticker/entity extraction and intent parsing; live market and web/news evidence retrieval channels.

5. Preprocessing Pipeline

Preprocessing is intentionally retrieval-first and deterministic to reduce hallucination risk and improve reproducibility:

Query Normalization: Canonicalization of user prompts to stabilize downstream retrieval behavior and reduce query variance.

Ticker and Entity Extraction: Symbol detection with alias mapping for robust instrument coverage, including support for major indices (^IXIC, ^GSPC, ^DJI) via a handcrafted Ticker Knowledge Base.

Query Expansion: Context-aware prompt reformulation to improve recall across search and news retrieval channels.

Evidence Filtering: Relevance-based filtering and deduplication of retrieved snippets before synthesis to minimize noise.

Context Packaging: Deterministic assembly of compact evidence blocks for Groq/SambaNova reasoning, ensuring token efficiency and faithful grounding.

6. Key Features

Mission Control UI	Premium dark-mode dashboard featuring glassmorphism design, glowing metric cards, and an intuitive conversational interface for financial querying.
Multi-Provider LPU Support	Toggle between Groq (Llama-3.3-70B-Versatile, DeepSeek-R1) and SambaNova (Llama-3.1-405B) for optimal latency/reasoning trade-offs.
Temporal Grounding	Real-time system context injection resolves relative date expressions accurately, preventing the temporal hallucination endemic to static LLMs.
Automatic Failover	Provider-level 429 error detection with automatic fallback between Groq and SambaNova for institutional-grade uptime.
Investment Compare Mode	Controlled side-by-side evaluation of two tickers across valuation, momentum, risk, and catalyst dimensions.
Raw Evidence View	Expandable retrieval payloads for transparent manual verification of all AI-generated claims.
Index Intelligence	Pre-mapped ticker knowledge base covering all major market indices (NASDAQ, S&P 500, Dow Jones, etc.).

7. Project Objectives

Primary Objective: Deliver a robust finance intelligence assistant that transforms natural-language prompts into verifiable, market-grounded analysis via multi-agent orchestration.

Reliability: Preserve source-grounded reasoning with deterministic retrieval and transparent evidence flow.

Usability: Provide a concise dashboard workflow for exploratory and comparative financial analysis.

Extensibility: Maintain modular architecture to integrate additional tools, data providers, and specialist agents.

Performance: Sustain low-latency execution suitable for iterative, analyst-in-the-loop decision making.

8. Technology Stack

Layer	Technology	Purpose
LLM Inference	Groq API / SambaNova	Sub-second reasoning LPU-accelerated
LLM Models	Llama-3.3-70B, DeepSeek-R1, Llama-3.1-405B	Versatile and elite financial synthesis
Agent Framework	phi (phidata)	Multi-agent orchestration
Finance Data	yfinance	Live price, fundamentals, news
Web Search	DuckDuckGo API	Real-time news and macro context
UI	Streamlit	Interactive dashboard
Persistence	SQLite (agents.db)	Agent memory and session history
Runtime	Python 3.9+	Core execution environment

9. Project Structure

streamlit_app.py — Main UI and orchestration logic — the 'Brain' of the application

finance_agent_team.py — Core agent definitions using the phi framework

run_all.sh — Automation script for environment setup and one-command launch

requirements.txt — Master list of Python dependencies

agents.db — Persistent SQLite database for agent memory and session history

.env.example — Environment variable template for API key configuration

10. Future Scope

Integration of SEC EDGAR filings for regulatory document analysis alongside live market data.

Portfolio-level risk aggregation across multiple tickers with volatility and correlation matrices.

Addition of a dedicated Quantitative Agent leveraging structured numerical reasoning models.

WebSocket-based real-time market feed for tick-by-tick data ingestion during market hours.

User authentication and persistent portfolio watchlists across sessions.

LangSmith or equivalent observability layer for production-grade agent trace monitoring.

